

Elijah Zhao

(306)-477-1048 | elijahzhao24@gmail.com | [linkedin.com/in/elijahzhao24](https://www.linkedin.com/in/elijahzhao24) | elijahzhao.com

EDUCATION

University of British Columbia

Bachelor of Science, Major in Computer Science, Minor in Math

GPA: 4.1/4.33

Expected Graduation: May 2028

TECHNICAL EXPERIENCE

Software Engineer Intern

May 2026 – Present

Hootsuite

Vancouver, BC

- Built an **Apache Calcite**-based **JDBC** connector for Hootsuite Analytics data, exposing analytics data as relational tables and using **filter pushdown** to optimize Tableau and Power BI SQL queries.
- Built the first Playwright **E2E** test suite for our MCP connector platform, covering flows across **3** services and mocking flaky MCP/OAuth dependencies in a **Docker** Compose environment; validated **10+** failure paths.
- Built a **Go** middleware to track Analytics API usage, streaming events through **Kafka** into Mixpanel for insights.

Software Engineer Intern

January 2026 – April 2026

Nexera Robotics (acquired by Locus Robotics)

Vancouver, BC

- Led redesign of a robotic pick-cycle pipeline in **Python/C++**, parallelizing ML inference and NumPy-based decision logic with **Ray Core** and optimizing critical paths in C++, improving throughput by **30%** (7→10 picks/min).
- Built internal robotics tooling in **Python/PostgreSQL** interfacing with **Universal Robots** firmware, shipping **5+** features including custom speed/pathing controls and performance tracking to identify system bottlenecks.
- Developed a [dataset curation pipeline](#) that reduced **1M+** unfiltered images to **7K** using **DINOv2 embeddings** and **pgvector** similarity search for deduplication and sampling; stored images in **S3** and metadata/vectors in **RDS**.

Software Development Lead

August 2025 – Present

UBC BizTech

Vancouver, BC

- Leading development of hackathon and conference platforms for UBC's largest tech club, used by **1,900+** users.
- Built a **serverless** backend with **AWS Lambda + Cognito** to manage authentication, payments, and event workflows; integrated **Stripe webhooks** to automate **\$20K+** in membership payments and event confirmations.
- Designed **DynamoDB** access patterns with **GSIs** enabling queries across **9+** events and **900+** members.
- Managed **AWS Lambda** infrastructure using **CloudFormation** (YAML), ensuring consistent dev/prod deploys.
- Built a [membership portal](#), [NFC networking system](#), and [mock stock exchange](#) with **TypeScript** and **Next.js**.

Undergraduate Teaching Assistant - CPSC 221

January 2026 – April 2026

University of British Columbia

Vancouver, BC

- Mentored **110+** students during lectures and weekly labs, teaching **data structures and algorithms** in **C++**.

Software Engineer Intern

May 2024 – August 2024

Woohelps International Technology LTD

Saskatoon, SK

- Built a **React/Django/PostgreSQL** apartment platform serving **~400** weekly users.
- Implemented **S3 presigned** uploads and WeChat OAuth authentication, used by **~80%** of platform users.

PROJECTS

[BizBot \(NwHacks 2026\)](#) | [Supabase](#), [YOLOv8](#), [FastAPI](#), [Next.js](#), [Google Gemini](#)

January 2026

- Won **Best Real-World Ready AI Product** among **630+** hackers by building an autonomous robot photographer with a **YOLOv8/DBSCAN** vision loop to auto-trigger photo capture and publish an **AI-filtered** live gallery.

[Failure Cloud \(Berkeley AI Hackathon 2026\)](#) | [FastAPI](#), [Next.js](#), [PyBullet](#), [NumPy](#)

May 2026

- Built a robot edge-case simulator that turns natural-language tasks into exportable **URDF/USD**-ready scenarios for robotics simulators; demoed in **PyBullet** with RGB/depth frames, point-cloud data, and failure evaluation.

[NBAsalary](#) | [Spring Boot](#), [JPA](#), [Python](#), [Java](#), [React](#), [TailwindCSS](#)

August 2025 – September 2025

- Built a full-stack CRUD app with a **Spring Boot** backend, modeling relational entities and using **Spring Data JPA** for ORM and custom queries on **PostgreSQL**; integrated with a **React** frontend to visualize salary cap data.

TECHNICAL SKILLS

Languages: C++, Python, Java, Go, C, JavaScript/TypeScript, SQL (PostgreSQL + pgvector)

Frameworks & ML: Spring Boot, Next.js, React, Node.js, FastAPI, Django, Ray, YOLOv8, OpenCV

Cloud & Tools: AWS (Lambda/DynamoDB/EC2/Amplify/Cognito/API Gateway), Supabase, Docker, Git, Linux, Kafka